From the Light Cone to Feynman's Path Sum at $\alpha = 2$

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One-sheet companion to *An Axiomatic Derivation of the Feynman Path Sum at the Light-Cone Kernel Width and Closure Order $\alpha = 2$ of the CAL*

Origin. CAL is not a theory of quantum gravity. It is an information-physics engine for non-Markovian reality, built to lift self-attention to a field-theoretic form from first principles of cognition, causality and intelligence. Its derivations of known physics are unit tests: a law of causal memory that cannot recover established physics has no business modeling minds or societies.

Definition 1 (Information). Information := distinguishability of physical states.

Remark 1 (Difference and observation). To distinguish physical states is to observe a difference between them. Change is a difference between states along an ordered evolution; thus change is temporally ordered distinguishability. Observation here denotes physical distinguishability, not a required conscious observer.

Definition 2 (Causality). Causality := change in physical state, including its rate of change.

From On Information Physics: From State Distinguishability to Causal Accumulation, Sec. 1.

The Causal Accumulation Law (On Information Physics, Eq. 61)

$$T^{\mu\nu a}{}_{\rho\sigma}(x) = \underbrace{\alpha_{\text{frac}}\, J^{\mu\nu a}{}_{\rho\sigma}(x)}_{\text{present source}} - (1-\alpha_{\text{frac}}) \int_{J^-(x)} \underbrace{k(\tau; x, x')}_{\text{history kernel}}\ \underbrace{\Pi^{\mu\nu}{}_{\rho\sigma|\alpha\beta}{}^{\gamma\delta}(x, x')}_{\text{slot coupling}}$$
$$\times \underbrace{P^{\alpha}{}_{\alpha'}(x,x')\, P^{\beta}{}_{\beta'}(x,x')\, P_{\gamma}{}^{\gamma'}(x,x')\, P_{\delta}{}^{\delta'}(x,x')}_{\text{spacetime transport of the past to } x} \times \underbrace{\Phi^{a}{}_{b}(x)\, P^{b}{}_{c'}(x,x')}_{\text{internal channel action}}\ J^{\alpha'\beta'c'}{}_{\gamma'\delta'}(x')\, d^4x'.$$

Read left to right: what the field is now (T) equals a share α_{frac} of the present source, minus a share $1-\alpha_{\text{frac}}$ of the past. Only the causal past $J^-(x)$ counts. Each past contribution is weighted by how long ago it happened (k), carried to the present point so it can be compared there (the four P factors), and rotated in its internal channel (ΦP). Below, $\alpha_{\text{frac}} = 2$, so the prefactor $-(1-\alpha)$ is $+1$; the sector is spinless and gauge-neutral, where the transports are the identity; and the memory is spread over the light cone.

Definitions and symbols

Sym.	Meaning	Type	Step
$T^{\mu\nu a}{}_{\rho\sigma}$	Accumulated output tensor	tensor field	CAL
$J^{\mu\nu a}{}_{\rho\sigma}$	Indexed source current	tensor field	CAL
$J^-(x)$, τ	Causal past of x; proper time	set; real	CAL
J	Carrier $\sum_a Z_a e_a$	$M_2(\mathbb{C})$	0
e_a	Basis $e_0 = I_2$, $e_j = -i\sigma_j$	$M_2(\mathbb{C})$	0
α_{frac}	Interaction depth; here 2	real	CAL
ν_τ	Light-cone profile, radius $c\tau$	measure	2
τ_c	Kernel lag scale (the only scale)	real > 0	2
K_α, m_α	Light-cone memory; its symbol	op.; fn.	2, 3

Sym.	Meaning	Type	Step
Π, P, Φ	Coupling; propagators; internal	tensors	CAL
$\Box$	Wave operator $\partial_t^2 - c^2\nabla^2$	operator	4
G, G_μ	Retarded Green's fns. ($\mu = 0$; μ)	kernels	4, 7
u, v	Null coordinates $ct \mp x$	real	4
μ, m	Rest rate $\mu\tau_c = 1/\sqrt{2}$; $mc^2 = \hbar\mu$	real > 0	5
Φ_{2n+1}	Checkerboard amplitude	complex	8
$\hat{H}$	$\sqrt{m^2c^4 - \hbar^2c^2\nabla^2}$	s.a. op.	10
K_F	Feynman's free kernel	kernel	11

0. The carrier

Each channel carries a Boltzmann-type decay weight and a Feynman-type phase weight; four channels assemble into one matrix.

$$Z_a = A_a e^{-H_a/\tau_E} + B_a e^{iS_a/\hbar_{\text{eff}}}, \qquad J = \textstyle\sum_{a=0}^{3} Z_a e_a \in M_2(\mathbb{C}).$$

1. The law at closure order two

The present counts twice and the whole past adds. A free carrier balances its accumulated past; a source is the imbalance.

$$T = 2J + K_2 J, \qquad T = 0 \text{ (free)}, \qquad T = S \text{ (sourced)}.$$

2. The light-cone kernel

At lag τ the profile is the light front of radius $c\tau$.

$$\hat{\nu}_\tau(k) = \frac{\sin(c|k|\tau)}{c|k|\tau}, \qquad K_\alpha f = \tau_c^{-\alpha} \int_0^\infty \frac{\tau^{\alpha-1}}{\Gamma(\alpha)} (\nu_\tau * f)(t-\tau)\, d\tau.$$

3. Lorentz invariance selects $\alpha = 2$ [Derived]

With $s = R\cos\varphi$, $c|k| = R\sin\varphi$:

$$\tau_c^{\alpha} m_\alpha = R^{-\alpha} \frac{\sin((\alpha-1)\varphi)}{(\alpha-1)\sin\varphi} = R^{-\alpha}\Big[1 + \frac{\alpha(2-\alpha)}{6}\varphi^2 + \dots\Big].$$

Lorentz invariant only at $\alpha = 2$ (and trivially 0).

4. The kernel is the retarded Green's function

$$G^{(3)} = \frac{\theta(t)\,\delta(|x| - ct)}{4\pi c^2 t}, \qquad G^{(1)} = \frac{\theta(u)\theta(v)}{2c} = \frac{1}{4c^2} W_u^1 W_v^1.$$

Kirchhoff's shell in $3+1$; in $1+1$, one order-one accumulation per null leg: $2 = 1+1$.

5. The balanced law is Klein–Gordon

$$T = 0 \implies (\Box + \mu^2) J = 0, \qquad \mu\tau_c = \tfrac{1}{\sqrt{2}}, \qquad mc^2 = \hbar\mu.$$

The phase belongs to the source: the double accumulation supplies the half turn; nothing multiplies the memory. Only mass ratios are physical.

6. Conservative exactly at $\alpha = 2$

$$\operatorname{Im}\omega = \frac{\pi}{4}(\alpha - 2)\frac{\mu^2}{|\omega|}.$$

Below 2 both signs of frequency decay; above 2 the causal response runs away. Nothing near 2 selects a sign.

7. The path sum [Derived]

The Neumann series of the balance converges absolutely and is a sum over piecewise-null histories, with $-\mu^2$ at each corner.

$$J = \tfrac{1}{2}\textstyle\sum_{n\geq 0}(-\mu^2)^n G^{*n} * S, \qquad G_\mu = \sum_{n\geq 0}(-\mu^2)^n G^{*(n+1)}.$$

8. Feynman's checkerboard in $1+1$

$$G^{*(n+1)} = \frac{\theta(u)\theta(v)}{2c}\frac{(uv/4c^2)^n}{(n!)^2}, \qquad (-\mu^2)^n G^{*(n+1)} = \frac{\Phi_{2n+1}}{2ic\mu}.$$

Term by term, because $(-\mu^2)^n = (i\mu)^{2n}$: each history of n insertions is a null zigzag with $2n+1$ corners.

9. The corners fill the cone in $3+1$

$$G_\mu = \frac{\delta(r-ct)}{4\pi cr} - \theta(ct-r)\frac{\mu}{4\pi c^2}\frac{J_1(\mu\sqrt{w}/c)}{\sqrt{w}}.$$

Histories without corners lie on the cone; the Bessel tail inside it is the corners alone (by descent).

10. Unitarity and the readout

$$J(t) = e^{-i\hat{H}t/\hbar} J(0), \qquad Q = \textstyle\int JJ^\dagger \text{ constant}.$$

Each sign of frequency is unitary; the sign is inherited from the source's past. On one sign the trace readout is Born-normalized. Markov on the pair $(J, \partial_t J)$.

11. Feynman's kernel

$$\frac{2mc^2}{i\hbar}\, e^{imc^2 t/\hbar}\, [G_\mu]^+(t) \longrightarrow K_F(t) = e^{i\hbar t\nabla^2/2m} \qquad (c \to \infty),$$

with error $O(1/c^2)$; its composition is Feynman's time slicing. The dynamics is real: the i of K_F enters only with the inherited sign of frequency.